DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-100-02 — Bench test relief valve 3"-P150-RV-4002/2 on T-101-N3

Area	A100 Tank farm	**Task**	PR Relief valve bench test and reinstatement
Priority	1 (safety)	**Issued by**	Facility owner (demonstration)
Primary asset	T-101 — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	3"-P150-RV-4002, 4"-P150-FL-4005
Isometrics	ISO-RV-4002, ISO-FL-4005 (owner's drawings, attached)		
Permits	General work permit, Lifting plan		

Reason

Relief valve due for its periodic bench test.

Scope of work

1. Confirm T-101 depressured and isolated.
2. Unbolt and remove VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET (3"-P150-RV-4002/2); blind the nozzle and the outlet line.
3. Bench test: set pressure 150 psi and seat tightness (SW-TEST-3); service the valve as found.
4. Reinstall with new gaskets and studs, tightened per SW-WO-5; remove blinds.

Affected items: 3"-P150-RV-4002/2, 3"-P150-RV-4002/1, 4"-P150-FL-4005/A1

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-3, SW-WO-1, SW-WO-2, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-RV-4002	3"-P150-RV-4002	3"-P150-RV-4002	1	0 error(s), 1 warning(s) — see below
ISO-FL-4005	4"-P150-FL-4005	4"-P150-FL-4005	3	0 error(s), 2 warning(s) — see below

ISO-RV-4002 (PDF): ../isometrics-imported/ISO-RV-4002.pdf · ISO-FL-4005 (PDF): ../isometrics-imported/ISO-FL-4005.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-RV-4002	1	warning	TO label 'PSV' is not on the drawing
ISO-FL-4005	3	warning	weld 4"-P150-FL-4005/W1 (E 22'-6" N 4'-6" EL 123'-6 11/16", sheet 3) has no label
ISO-FL-4005	1	warning	FROM label 'PSV ON T-101 NOZ. N3' is not on the drawing

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
3"-P150-RV-4002/2	item	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 124'-0"
3"-P150-RV-4002/1	item	GASKET 3" P150 COMPRESSED SHEET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 123'-4-3/8"
4"-P150-FL-4005/A1	item	GASKET 4" P150 COMPRESSED SHEET	4"-P150-FL-4005	E 22'-0" N 25'-10-1/16" EL 124'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	T-101	3"-P150-RV-4002	T-101 depressured, drained and isolated under the shutdown isolation (no inlet valve)			
2	blind	3"-P150-RV-4002/1	3"-P150-RV-4002	Blind the nozzle T-101-N3 once the valve is removed	E 22'-0" N 26'-6" EL 123'-4-5/16"		
3	blind	4"-P150-FL-4005/A1	4"-P150-FL-4005	Blind the outlet line 4"-P150-FL-4005 at its first flange	E 22'-0" N 25'-10-1/8" EL 124'-0"		

Parts

Description	Qty	Replaces
GASKET 3" P150 COMPRESSED SHEET	1	3"-P150-RV-4002/1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P150	1	3"-P150-RV-4002/3
GASKET 4" P150 COMPRESSED SHEET	1	4"-P150-FL-4005/A1

Description	Qty	Replaces
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1	4"-P150-FL-4005/A3

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
3"-P150-RV-4002/1	4 × 0.875" studs (3"-P150-RV-4002/3)	27,059	3	
4"-P150-FL-4005/A1	4 × 1" studs (4"-P150-FL-4005/A3)	35,343	3	

Test — Relief valve bench test

Description	Set pressure 150 psi (class design pressure, PSV-1) and seat tightness
Pressure	150 psi
Basis	SW-TEST-3

Inspection and test plan (SW-ITP-1..3)

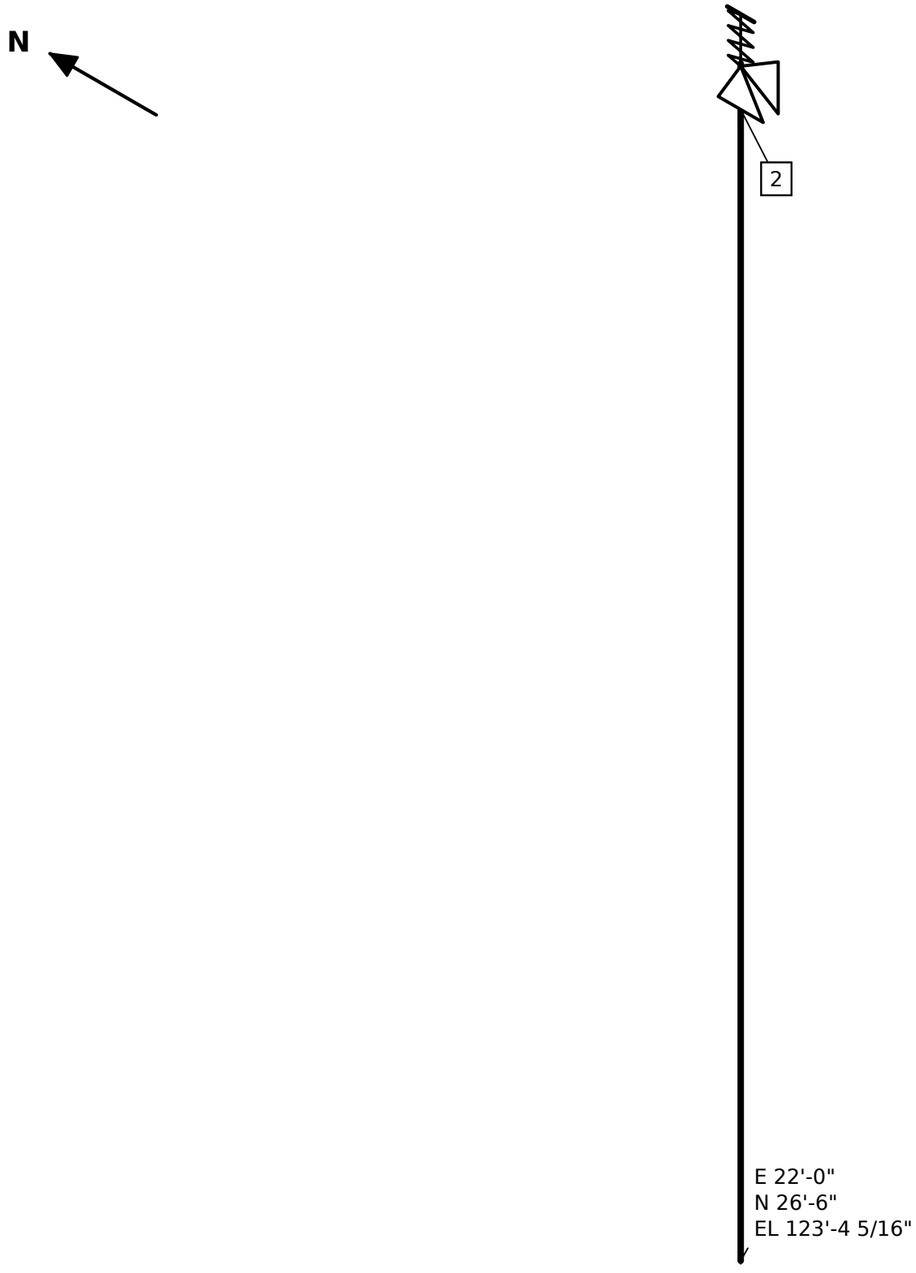
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Relief valve removed; nozzle and outlet blinded	SW-WO-3	Both openings blinded and tagged	Isolation certificate	W	W	—				
3	As-found bench test (pop pressure recorded before any service)	SW-TEST-3	Recorded	Bench test record	W	W	W				
4	Set pressure and seat tightness after service	SW-TEST-3, PSV-1	Set at 150 psi; seat tight	Bench test record, seal tag	H	H	W				
5	Valve reinstalled; new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
3"-P150-RV-4002/2	affected	ISO-RV-4002	1	
3"-P150-RV-4002/1	affected	ISO-RV-4002	1	
4"-P150-FL-4005/A1	affected	ISO-FL-4005	1	
3"-P150-RV-4002/1	isolation: blind	ISO-RV-4002	1	
4"-P150-FL-4005/A1	isolation: blind	ISO-FL-4005	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	3"	GASKET 3" P150 COMPRESSED SHEET
2	1	3"x4"	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET
3	1	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P150

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. PAINT PER PROJECT SPECIFICATION

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

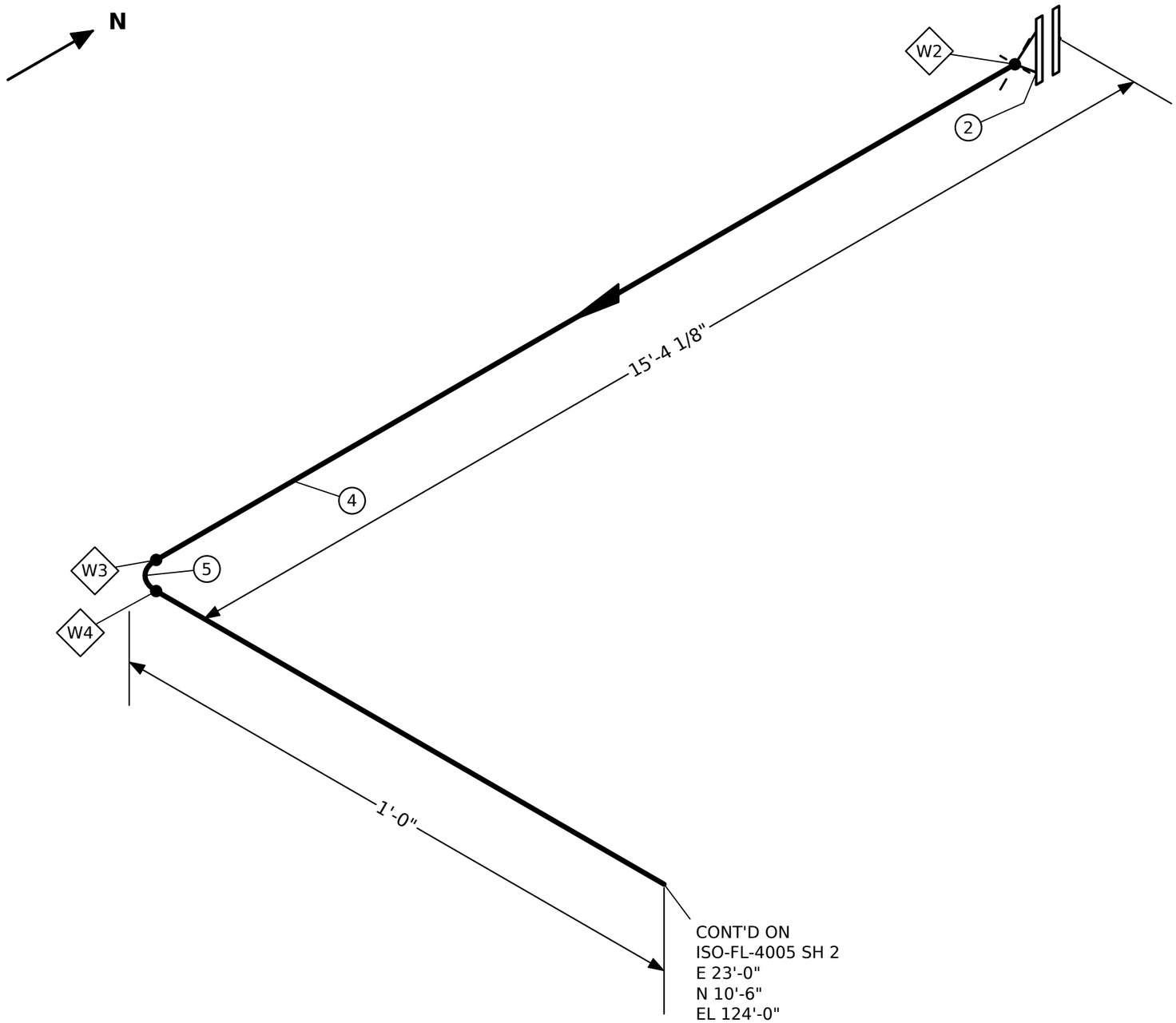
SMARTWELD

TITLE

HYDROCARBON PROCESS

TEST PRESS. -	INSULATION NONE	PAINT -	
PWHT -	NDE -	FLUID RV	

LINE NO. 3"-P150-RV-4002	PIPING CLASS H150	DESIGN PRESS. 150 PSIG	DESIGN TEMP. -
DRAWING NO. ISO-RV-4002		SHEET 1 OF 1	REV 0
DATE 2026-11-01	DRAWN -	CHECKED -	SCALE NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	4"	GASKET 4" P150 COMPRESSED SHEET
2	1	4"	FLANGE WELD-NECK 4" P150 RAISED FACE
3	1	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
4	14'-9 1/4"	4"	PIPE 4" P150 BEVELLED ENDS
5	1	4"	ELBOW 90° R = 5/3 OD 4" P150

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	FL

COMPANY

SMARTWELD

TITLE

FLARE AND RELIEF OUTLETS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P150-FL-4005	F150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-FL-4005	1 OF 3	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

Notices

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

This document embeds a subset of the DejaVu Sans font (SW-LIC-3), renamed under its licence. DejaVu Sans (assets/fonts/DejaVuSans.ttf), from the DejaVu fonts 2.37 as packaged in the build environment.

Copyright (c) 2003 by Bitstream, Inc. All Rights Reserved. Bitstream Vera is a trademark of Bitstream, Inc.
DejaVu changes are in public domain.

Permission is hereby granted, free of charge, to any person obtaining a copy
of the fonts accompanying this license ("Fonts") and associated
documentation files (the "Font Software"), to reproduce and distribute the
Font Software, including without limitation the rights to use, copy, merge,
publish, distribute, and/or sell copies of the Font Software, and to permit
persons to whom the Font Software is furnished to do so, subject to the
following conditions:

The above copyright and trademark notices and this permission notice shall
be included in all copies of one or more of the Font Software typefaces.

The Font Software may be modified, altered, or added to, and in particular
the designs of glyphs or characters in the Fonts may be modified and
additional glyphs or characters may be added to the Fonts, only if the fonts
are renamed to names not containing either the words "Bitstream" or the word
"Vera".

This License becomes null and void to the extent applicable to Fonts or Font
Software that has been modified and is distributed under the "Bitstream
Vera" names.

The Font Software may be sold as part of a larger software package but no
copy of one or more of the Font Software typefaces may be sold by itself.

THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS
OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT,
TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL BITSTREAM OR THE GNOME
FOUNDATION BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING
ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES,
WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF
THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE
FONT SOFTWARE.

Except as contained in this notice, the names of Gnome, the Gnome
Foundation, and Bitstream Inc., shall not be used in advertising or
otherwise to promote the sale, use or other dealings in this Font Software
without prior written authorization from the Gnome Foundation or Bitstream
Inc., respectively. For further information, contact: fonts at gnome dot
org.